

Reduced row echelon formReduced row echelon form:

row echelon form

+ all pivots = 1

+ entries above pivots = 0hence entries above and below the pivots are zero

To turn the matrix from row echelon form to reduced row echelon form, we carry on with elementary row operations:

Gauss-Jordan reduction

obtained last time:

$$\left(\begin{array}{ccccc|c} \textcircled{1} & 2 & 2 & 2 & 2 & 2 \\ 0 & 0 & \textcircled{2} & 4 & 4 & 1 \\ 0 & 0 & 0 & \textcircled{5} & 5 & 9 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) \quad \begin{array}{l} R_2 \rightarrow \frac{1}{2}R_2 \\ R_3 \rightarrow \frac{1}{5}R_3 \end{array}$$

row echelon form

$$\left(\begin{array}{ccccc|c} \textcircled{1} & 2 & 2 & 2 & 2 & 2 \\ 0 & 0 & \textcircled{1} & 2 & 2 & 0.5 \\ 0 & 0 & 0 & \textcircled{1} & 1 & 1.8 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) \quad \begin{array}{l} R_1 \rightarrow R_1 - 2R_3 \\ R_2 \rightarrow R_2 - 2R_3 \end{array}$$

$$\left(\begin{array}{ccccc|c} \textcircled{1} & 2 & 2 & 0 & 0 & -1.6 \\ 0 & 0 & \textcircled{1} & 0 & 0 & -3.1 \\ 0 & 0 & 0 & \textcircled{1} & 1 & 1.8 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) \quad R_1 \rightarrow R_1 - 2R_2$$

$$\left(\begin{array}{ccccc|c} \textcircled{1} & 2 & 0 & 0 & 0 & 4.6 \\ 0 & 0 & \textcircled{1} & 0 & 0 & -3.1 \\ 0 & 0 & 0 & \textcircled{1} & 1 & 1.8 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) \quad \text{reduced row echelon form}$$

From the reduced row echelon form, it is easier to get the solution set:

$$\begin{array}{l} x_2 = s \\ \xrightarrow{\hspace{1cm}} \\ x_5 = t \\ \xrightarrow{\hspace{1cm}} \end{array}$$

$$x_4 + t = 1.8$$

$$\text{Hence } \underline{x_4 = 1.8 - t} \rightarrow$$

$$\underline{x_3 = -3.1} \rightarrow$$

$$x_1 + 2s = 4.6$$

$$\text{Hence } x_1 = 4.6 - 2s$$

Octave/matlab: `rref(A)`

Remark: The reduced row echelon form is unique.

Inverses of Matrices

real numbers

multiplicative identity:

1

multiplicative inverse of x :

is the number y such that

$$x \cdot y = y \cdot x = 1$$

(Hence the inverse is $1/x$, which is also denoted x^{-1} .)

square matrices

multiplicative identity:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \text{ etc}$$

$I_2 \qquad I_3 \qquad I_4$

multiplicative inverse of square matrix A :

is any square matrix B such that

$$A \cdot B = B \cdot A = I$$

identity matrix of
suitable size

Definition Let A be an $n \times n$ matrix. Then an $n \times n$ matrix B is an inverse of A if

$$AB = BA = I_n.$$

If A has an inverse, we say A is invertible.

Example Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$.

Then $\begin{pmatrix} 1 & 0 \\ 0 & 1/2 \end{pmatrix}$ is an inverse of A .

Check:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1/2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

I

$$\begin{pmatrix} 1 & 0 \\ 0 & 1/2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

I

Example Let $A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} ?? & \\ & \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

A is not invertible.

Example Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \cdot & \cdot \\ \cdot & \cdot \end{pmatrix} = \begin{pmatrix} \cdot & \cdot \\ 0 & 0 \end{pmatrix} \neq I$$

A is not invertible.

Example $\begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}$ is also not invertible.
Check this!